

Dhruv Nigam

MACHINE LEARNING ENGINEER

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Work Experience

Dream 11

Mumbai

LEAD MACHINE LEARNING ENGINEER

July 2022 - Present

- Trained **200M-parameter cross-attention transformer (Lumos)** on **180B user tokens** from **250M users**; ran **daily batch inference for 70M users** across **200 L4 GPUs**. Replaced **50+ task-specific models**. **+3.15% DAU**. Built **Accio**, a Delta-native streaming data loader for distributed training on **8x A100s**. [\[tech report\]](#), [\[blog\]](#)
- Post-trained Llama 8B with GRPO (RLHF)** for a sports banter bot. Trained a **DeBERTa-v3 reward model** on **222K preference pairs** to score response engagingness. Served via **vLLM on A100** at **3,000 RPM**, **400ms TTFT**. **+16% feature retention**.
- Built **real-time AI voice agent** for live cricket streams: **speech-to-speech over WebRTC**, **sub-2s latency**, scaled to **10K concurrent sessions**. **1.5M min watchtime** (IPL 2026). [\[blog\]](#)
- Built **Pacman**, a **personalized promotion platform** for **200M users** allocating **\$90M+ annually**. **+14% WAU**, **85% ROI**. Extended it with **RL**: framed discounting as a **budget-constrained MDP**, built a data-driven simulator from production logs, trained **Q-learning/DQN agents (40M+ episodes)**, deployed in a **live A/B across 2.2M users**.
- Developed **ForeCal**, a probability calibration technique, **published at KDD 2024** [\[paper\]](#). Improved accuracy of churn prediction model used for offer targeting. Reduced calibration error by **50%**, **0.4% revenue upside**.

Protium

Mumbai

DIRECTOR - DATA SCIENCE AND ENGINEERING

Feb. 2020 - June 2022

- 12th employee and **founding ML Engineer**. Automated the **end-to-end credit underwriting workflow**: deep learning credit models on **5M+ user time series (+0.12 AUC vs bureau scores)**, **Bayesian income estimation**, and **rule engines** with deviation-based human escalation. **\$500M+ in loans** underwritten **straight-through**. [\[blog\]](#)
- Built the **ML platform** that hosted 52+ models as FastAPI endpoints for **real-time feature engineering and inference** supporting up to **10,000 rpm**.

CLSA

Mumbai

QUANTITATIVE TRADER

Nov. 2017 - Feb. 2020

- Founding member of a 3-person ML team. Built **model-based trade execution strategies** handling **\$1 bn/day** in client trades at **sub-10ms latency**. Developed a **Java-based event-driven system** for real-time feature calculation and low-latency ML inference.
- Developed a profitable quantitative trading strategy (**Sharpe 1.3**) using **cointegration-based time series statistical arbitrage** in US futures. Traded over 2 years with **\$10M in capital**, **\$4M net profit**.

Open Source, Talks & Writing

Talks ForeCal at **KDD 2024** (Barcelona), LUMOS at **Swiggy** (invited), GPU Fundamentals at **MumbaiPy**. [\[all talks\]](#)

Writing Articles on mltrenches.substack.com: Algorithmic Intensity, RL from Scratch, Voice Agent Lessons Learned.

Open Source Merged PRs in [LiveKit Agents](#) and [TRL](#) (HuggingFace).

Education

Indian Institute of Technology - Bombay (IITB)

Mumbai

BACHELORS AND MASTERS IN ELECTRICAL ENGINEERING, SPECIALIZING IN SIGNAL PROCESSING. THESIS: [Single Channel Blind Dereverberation of Speech Signals](#)

2011 - 2016

Technical Skills

LLMs & Post-Training	LLM post-training (SFT, GRPO, RLHF), reward modeling, reinforcement learning, TRL, vLLM, PyTorch
Agents	Agentic workflows, LangGraph, LiveKit Agents, real-time voice agents (WebRTC, speech-to-speech)
Data & Infra	Apache Spark, PySpark, Databricks, Delta Lake, Apache Kafka, Apache Flink, Pandas, Scikit-learn
Cloud & MLOps	AWS (EMR, S3, Glue, Lambda), GCP (BigQuery, Cloud Run, GKE), Docker, Kubernetes, FastAPI, MLflow, Ray